Keysight PROPSIM

Multi-Emulator Synchronisation

Application Note

KEYSIGHT TECHNOLOGIES

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1 INTRODUCTION

Multi-emulator synchronization enables applications that go beyond the capability of a single emulator, like setups with high number of RF inputs or outputs in:

- MIMO OTA testing
- Multi-user MIMO testing
- Network level testing

The multi-emulator setup principle is illustrated in Figure 1. Synchronization is done by connecting the time base of all emulators to same external 10 MHz reference signal. The emulation is started simultaneously in all emulators with trigger signal. Emulator 1 generates the trigger for the emulator 2.

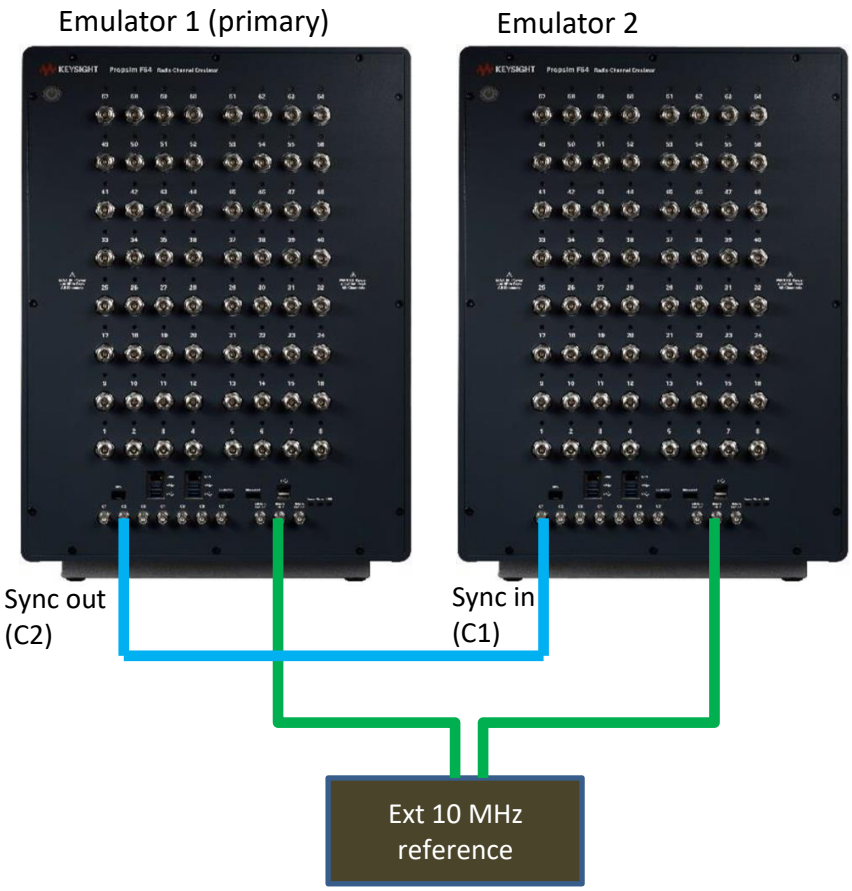

Figure 1. Multi-emulator setup

Two PROPSIM F64 devices can be used in the setup. The devices can be controlled either with GUI of each device or with remote control from external PC.

1.1 Document history

The following table lists the main changes to issues of this document.

Issue	Date	Summary of Changes
1.0		Release of PROPSIM Multi-Emulator Sync
1.1	Dec 2014	New documentation template
1.2	Jan 2016	New documentation template
1.3	Jan 2018	New documentation template

Issue	Date	Summary of Changes
1.4	Jun 2019	PROPSIM F64 changes
2.0	Feb 2020	PROPSIM GUI changes.
2.1	Mar 2021	Terminology changes.
2.2	Jun 2021	Customer support information updated.

2 PERFORMANCE, FEATURES & SETUP REQUIREMENTS

The multi-emulator performance and features are described in Table 1. Also the setup requirements are listed.

Table 1. Multi-emulator performance, features and setup requirements

Parameter	Value
PROPSIM HW	All the emulators of the setup shall be of equal model, firmware and HW version, and calibrated. Contact Keysight for further information.
PROPSIM firmware	PROPSIM F64: Release 1.0 or above
Synchronization accuracy	+/- 80 ns
Number of emulators	2
Supported connection topologies	The same connection topology shall be used in all emulators of multi-emulator setup
Synchronization cables (between cascaded emulators)	Cable type: 5 ns / m (velocity factor 0.66) Connector type: BNC All the synchronization cables used in the setup shall be of equal length. Supported lengths: 2.0 m ± 0.02 m 4.0 m ± 0.02 m 6.0 m ± 0.02 m
10 MHz clock cables	Connector type: BNC All the reference cables used in the setup shall be of equal type and length.
External 10 MHz ref clock	Connector type: BNC Impedance: 50 ohm Signal level: > 0 dBm, +20 dBm max, >10 dBm for optimal performance Point to point connection to each emulator Duty cycle: 50 ± 2 % Max frequency deviation: ±5 Hz Timing accuracy between emulators: < 200 ps Phase accuracy between emulators: <2 degrees

Figure 2 and Figure 3 illustrate the synchronization (simultaneous start of emulations) with test setup of three emulators and 4.0 m long synchronization cables.

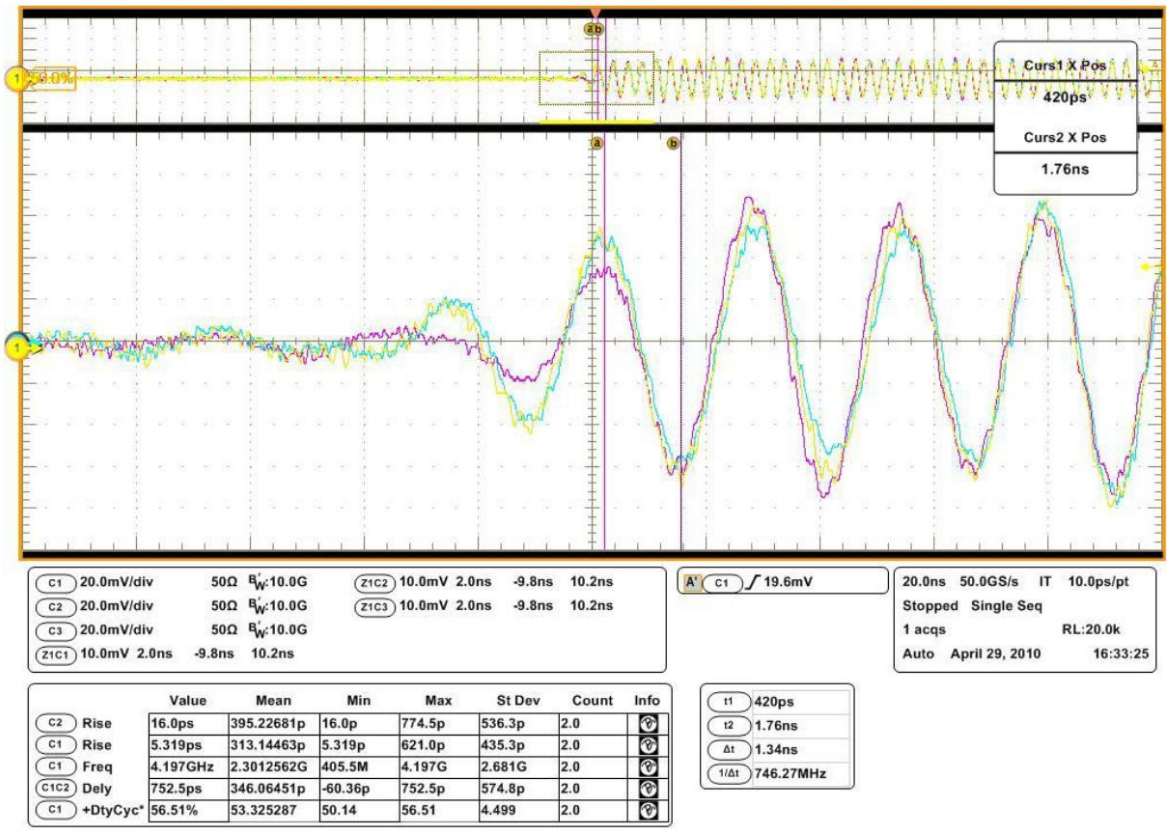

Figure 2. Synchronization with 3 emulators (example 1)

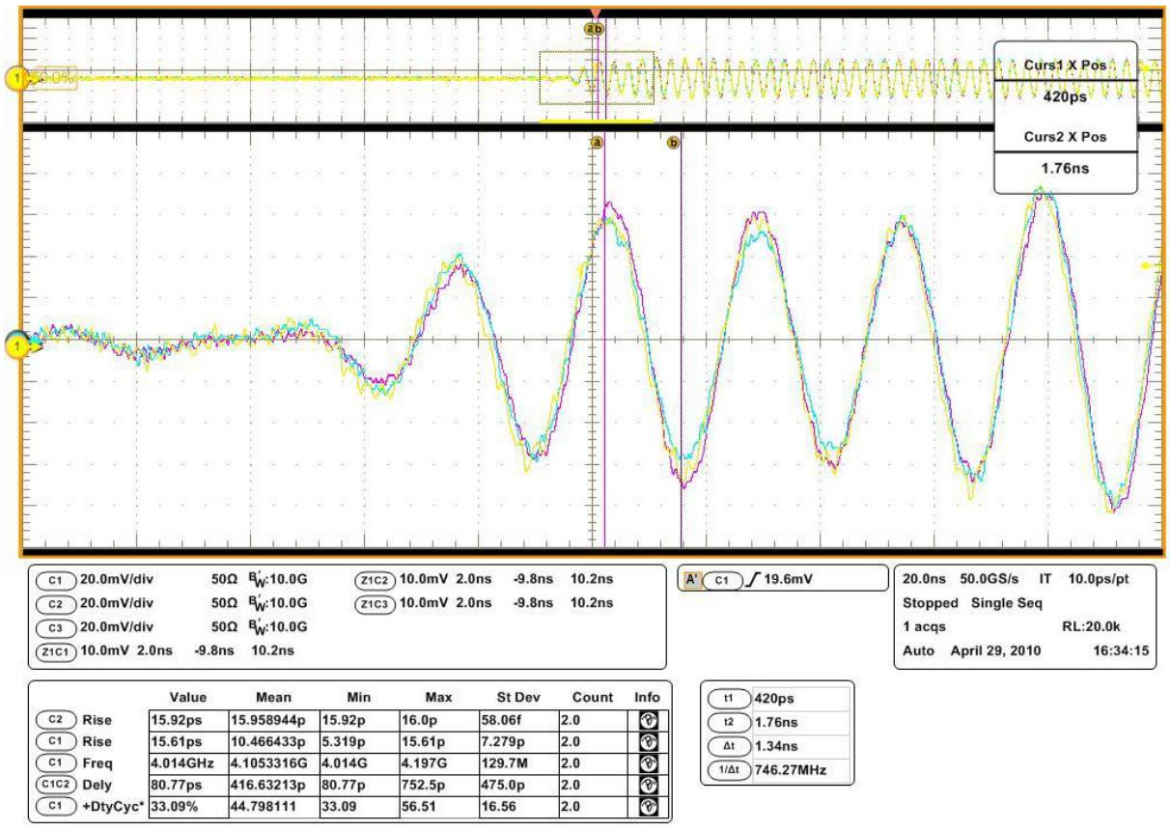

Figure 3. Synchronization with 3 emulators (example 2)

3 MULTI-EMULATOR SETUP

The connections in multi-emulator setup are illustrated in Figure 4. Note that the cable length for

- each synchronization shall be equal
- each 10 MHz reference clock connection shall be equal

See Table 1 for more detailed cabling requirements.

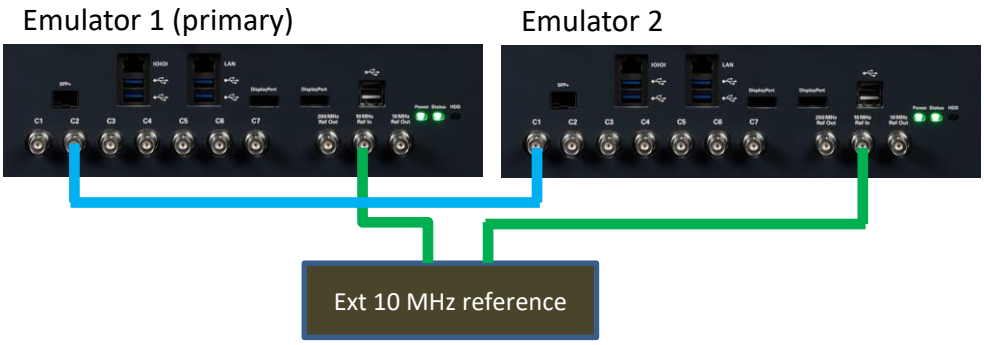

Figure 4. Multi-emulator setup connections

4 GENERATION OF CHANNEL MODELS

A separate channel model shall be crated for each emulator in the multi-emulator setup. As an example, GCM tool supports creating individual emulations automatically for multiple emulators.

The emulation files must be saved to the hard drive of each emulator. It is also possible to run models directly from network drive.

Note that it is essential to preserve the relative gains between different channel models both in the modelling phase and when running the emulation.

5 RUNNING EMULATIONS

The emulations can be started simultaneously in the multi-emulator setup according to instructions below. Either Remote control or Graphical User Interface can be used. After the emulations have been started synchronously, the green LEDs indicating active connectors starts blinking in each emulator. LED blinking is not synchronized between emulators, only the start of emulation running is synchronized.

5.1 Remote control (ATE)

1. Transfer all emulations to corresponding emulators
2. Enable external 10 MHz ref clock **in each emulator**
   - ROUTe:PATH:REFerence EXT
3. Configure multi-emulator setup **for each emulator**
   - SYSTem:MSIMulator:CABle <cable length>
   - SYSTem:MSIMulator:CONFig <emulator position>,<emulator total amount>

     Example: commands to 1st emulator in setup of 2 emulators with 4.0 meter long synchronization cable
     - SYSTem:MSIMulator:CABle 4.0 m
     - SYSTem:MSIMulator:CONFig 1,2
4. Open emulation **in each emulator**. Note: the emulations need to be reloaded, if the cable length or multi-emulator setup is changed.
   - CALCulate:FILTer:FILE <filename>
5. Set emulation parameters **in each emulator**
6. Set **emulator 2** to trigger on falling edge of trigger signal, enable triggering and start emulation
   - DIAGnostic:SIMUlation:TRIG:CONF EDGE,FALLING
   - DIAGnostic:SIMUlation:TRIG:SET ON
   - DIAGnostic:SIMUlation:GO
7. Start emulation **in emulator 1**
   - DIAGnostic:SIMUlation:GO

5.2 Graphical user interface

1. Transfer all emulations to corresponding emulators.
2. Enable external 10 MHz ref clock **in each emulator.**
   - Click the **INT REF /EXT REF** button in the status bar at the bottom of the PROPSIM GUI.
   - Select **EXT REF**.

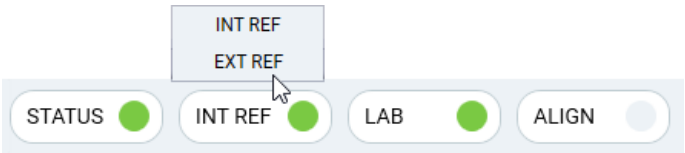

Figure 5. Selecting external reference

3. Configure multi-emulator setup **for each emulator.**
   - Select **Configuration** > **Device Configuration** in the navigation bar, see Figure 6.
   - Set length of synchronization cables and position of the emulator in the setup.

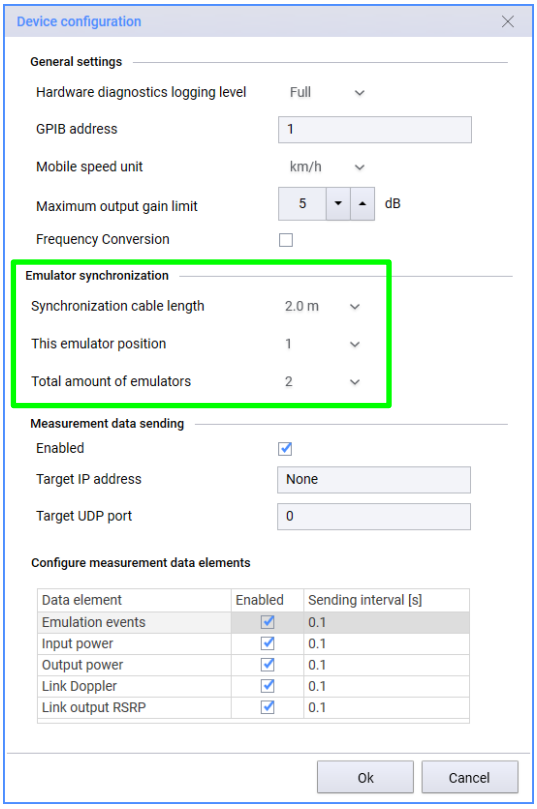

Figure 6. Device configuration dialog

4. Load emulations **to each emulator**. Note: the emulations need to be reloaded if the cable length or multi-emulator setup is changed.
   - Select **Emulation** > **Open** in the navigation bar, select the emulation file to be loaded, and click **Run**.
5. Setup emulation parameters **in each emulator.**
6. Set **emulator 2** to trigger on falling edge of trigger signal, enable triggering and start emulation.
   - In the emulation settings pane of the Emulation Control View, select the **External triggered emulation control** checkbox.
   - In the **External trigger active state** list, select "Falling Edge", see Figure 7. The background of the (Start) button starts blinking and the tooltip shows the text "Waiting for external signal".

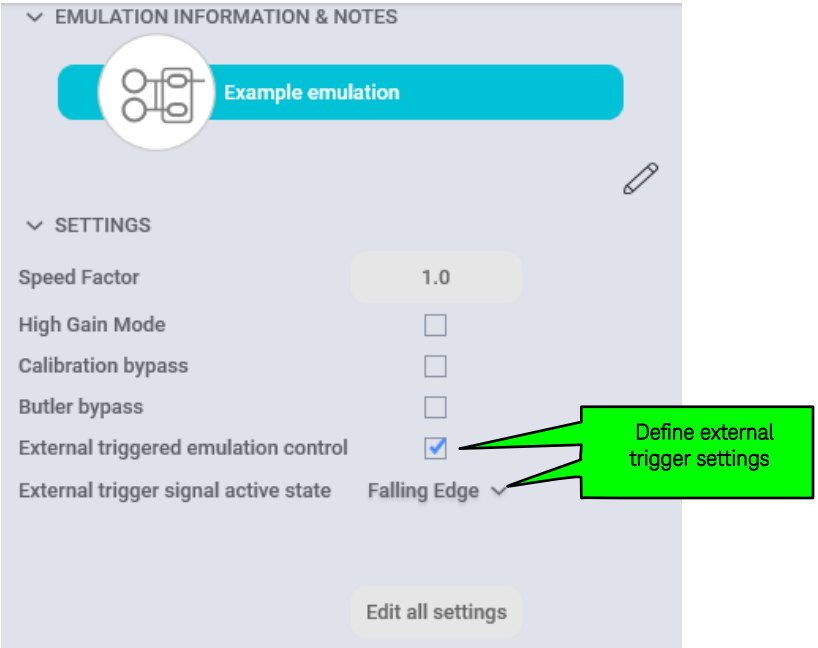

Figure 7. Emulation settings pane in Emulation Control View

7. Start emulation in emulator 1.
   - Click the (Start) button in the Emulation Control View.